

Sudarshan Aryal

Kathmandu, Nepal

Email: sudarshanaryal99@gmail.com | Phone: +91-9381285688

LinkedIn: <https://www.linkedin.com/in/sudarshan-aryal-94838a2a5/>

GitHub: <https://github.com/Sudu-09-Nep>

Substack: <https://sudarshanaryal.substack.com/>

Profile

Undergraduate student of Biosciences and Biotechnology with a growing interest in genomics, bioinformatics, and computational biology. I enjoy learning through hands-on projects, combining wet-lab biological concepts with data-driven analysis, and documenting my scientific thinking through writing and open-source work.

Education

BS (Hons.) with Research in Biosciences and Biotechnology

Sri Sathya Sai Institute of Higher Learning

2023 – 2027

CGPA: 8.68 (up to Semester 5)

Senior Secondary Examination

St. Xavier's College

2020 – 2023

GPA: 3.21 / 4.0

Projects & Research Experience

Hybrid Genome Assembly using Oxford Nanopore and Illumina Data

Mentored Bioinformatics Project

- Implemented hybrid genome assembly and polishing workflows using ONT and Illumina sequencing data. - Worked with tools including Filtlong, Autocycler, Polypolish, PyPolca, Roary, and Snippy. - Developed foundational understanding of microbial genomics, assembly strategies, and variant analysis. - GitHub: <https://github.com/Sudu-09-Nep/ont-illumina-hybrid-assembly>

Molecular Docking and Protein-Ligand Interaction Analysis

Learning-based Computational Project

- Practiced protein-ligand docking workflows using AutoDock Tools. - Prepared and analyzed molecular structures using UCSF Chimera and Open Babel. - Built conceptual understanding of structure-based molecular analysis through repeated hands-on practice.

Workshops & Training

Finishing School in Foundations of Cell Culture and Soft Skills

10-day Hands-on Workshop

Organized by Federation of Asian Biotechnologists Associations (FABA)

In association with Thermo Fisher Scientific

Technical Skills

Bioinformatics & Genomics: Genome assembly, genome polishing, pangenome analysis, variant calling, BLAST analysis

Computational Tools: Filtlong, Autocycler, Polypolish, PyPolca, Roary, Snippy, AutoDock, UCSF Chimera, Open Babel

Programming & Data Analysis: Python (basic), R (basic), Biostatistics

Laboratory Techniques: DNA extraction, PCR, agarose gel electrophoresis, SDS-PAGE, microbiology techniques

Computing Environment: macOS, Conda environments

Research Interests

- Microbial genomics and genome assembly
 - Gene expression and regulation
 - Bioinformatics applications for underrepresented populations
-

Writing & Science Communication

- Author of bioscience and research-learning essays on Substack
 - Interested in communicating scientific ideas through clear and reflective writing
-

Languages

Nepali (Native)

English (Fluent)

Hindi (Fluent)

Last updated: 2026